PRIVATE CYBER LAB HANDBOOK — DOCUMENT 03

# Lab 03: Organizational Unit (OU) Management

Active Directory Hierarchy, Group Policy Inheritance, and Safety Controls

**Operating System:** Windows Server 2019
**Active Directory Domain:** zroots.local
**Environment:** Windows PowerShell on DC01
**Domain Controller:** DC01 (10.1.1.10)
**Virtualization Platform:** VMware (Host-only Network)
**Target Audience:** Administrators & Portfolio Reviewers

## 2. Executive Summary

This report documents **Lab 03** of the Private Cyber Lab Handbook, focusing on Active Directory (AD) Organizational Unit (OU) design, hierarchy inspection, distinguished name (DN) parsing, computer account placement verification, Group Policy inheritance, and directory safety mechanisms.

Using Windows PowerShell on domain controller `DC01` within the `zroots.local` domain, we verified a twelve-OU hierarchical architecture. Special attention was given to validating default accidental-deletion protection across critical containers, identifying an un-protected state on the `Domain Controllers` OU, successfully remediating it via PowerShell, and analyzing GPO inheritance links for both Windows 10 and Linux environments.

## 3. Lab Objectives

- Understand Active Directory Organizational Units and their structural purpose.
- Inspect the complete OU hierarchy in the `zroots.local` domain.
- Comprehend distinguished names (DNs) and OU nesting mechanics.
- Verify the exact placement of Windows 10 and Windows 11 computer accounts.
- Inspect the Linux OU and its Group Policy inheritance parameters.
- Analyze Group Policy links and inheritance inheritance chains.
- Verify the Windows 10 Security Policy Group Policy Object (GPO).
- Understand accidental-deletion protection and directory safety.
- Enable and verify accidental-deletion protection on the `Domain Controllers` OU.
- Document commands, actual observed results, security implications, and troubleshooting.

## 4. Lab Environment and Topology

This lab utilizes the previously established isolated VMware environment (`MyLabNet` - `10.1.1.0/24`). No new network configuration or disruptive hardware modifications were performed during this lab exercise.

```
Internet
|
Physical Router — 192.168.0.1
|
VMware VMnet0 — Bridged Network
|
DC01 — Windows Server 2019 (10.1.1.10)
|-- Active Directory Domain Services (zroots.local)
|-- DNS & RRAS/NAT
|
MyLabNet — VMware Host-only Network (10.1.1.0/24)
|-- Windows 10: 10.1.1.20 | Windows 11: 10.1.1.50
|-- Kali Linux: 10.1.1.30 | BlackArch: 10.1.1.40 | Metasploitable: 10.1.1.60
```

`DC01` acts as the authoritative domain controller, hosting AD DS and DNS for `zroots.local`. Client VMs are domain-joined to this directory structure.

## 6. Active Directory OU Concepts

Organizational Units (OUs) are containers within an Active Directory domain that can hold users, groups, computers, and other OUs. They form the basis for administrative delegation and Group Policy Object (GPO) scope targeting.

**Delegation of Control:** OUs allow administrators to delegate permissions over specific subsets of users or machines without granting domain-wide administrative privileges.
**Policy Targeting:** GPOs are linked to domains, sites, or OUs, ensuring security baselines are applied selectively to specific organizational groups.
**Nesting:** OUs can be nested within other OUs to mirror real-world corporate hierarchies and business units.

## 7. OU Hierarchy and Inventory

The `zroots.local` domain contains 12 verified OUs. Initial inspection revealed that 11 OUs had accidental-deletion protection enabled by default, while the `Domain Controllers` OU required remediation.

| OU Name | Distinguished Name (DN) | Protected |
|---|---|---|
| Administrators | OU=Administrators,OU=User Accounts,DC=zroots,DC=local | True |
| Computer Accounts | OU=Computer Accounts,DC=zroots,DC=local | True |
| Domain Controllers | OU=Domain Controllers,DC=zroots,DC=local | True* |
| Groups | OU=Groups,DC=zroots,DC=local | True |
| Linux | OU=Linux,OU=Computer Accounts,DC=zroots,DC=local | True |
| Servers | OU=Servers,DC=zroots,DC=local | True |
| Service Accounts | OU=Service Accounts,DC=zroots,DC=local | True |
| Staff | OU=Staff,OU=User Accounts,DC=zroots,DC=local | True |
| Students | OU=Students,OU=User Accounts,DC=zroots,DC=local | True |
| User Accounts | OU=User Accounts,DC=zroots,DC=local | True |
| Windows 10 | OU=Windows 10,OU=Computer Accounts,DC=zroots,DC=local | True |
| Windows 11 | OU=Windows 11,OU=Computer Accounts,DC=zroots,DC=local | True |

*Note: Domain Controllers initially reported `False` and was subsequently updated to `True`.

## 8. Computer-Account Placement

Properly placing computer accounts in dedicated OUs (such as `OU=Windows 10` and `OU=Windows 11` under `OU=Computer Accounts`) is critical for precise Group Policy targeting, auditing, and organizational hygiene. Placing machines in the default `CN=Computers` container bypasses custom OU-level security baselines.

## 9. PowerShell Execution: Test 1

### Test 1: Inspect OU Hierarchy and Protection Settings

This command queries all OUs in the domain, sorts them by Distinguished Name, and evaluates the protection status using a calculated property.

```
Get-ADOrganizationalUnit -Filter * -SearchBase "DC=zroots,DC=local" |
Sort-Object DistinguishedName |
Select-Object Name, DistinguishedName,
@{Name="Protected";Expression={
(Get-ADOrganizationalUnit -Identity $_.DistinguishedName -Properties
ProtectedFromAccidentalDeletion).ProtectedFromAccidentalDeletion
}} |
Format-Table -AutoSize
```

> **Actual Findings:** Twelve OUs were returned. Eleven OUs showed `Protected=True` in the initial inventory. The `Domain Controllers` OU initially showed `Protected=False`, which was subsequently remediated and verified.

## 10. Group Policy Inheritance (Tests 2 & 3)

### Test 2: Inspect Linux OU Group Policy Inheritance

```
Get-GPInheritance -Target "OU=Linux,OU=Computer Accounts,DC=zroots,DC=local"
```

```
Name: linux | ContainerType: OU | Path: ou=linux,ou=computer accounts,dc=zroots,dc=local
GpoInheritanceBlocked: No | GpoLinks: {} | InheritedGpoLinks: {Default Domain Policy}
```

*Note:* This inspects policy inheritance for the container; it does not imply Windows GPOs successfully execute on Linux endpoints.

### Test 3: Inspect Windows 10 OU Group Policy Inheritance

```
Get-GPInheritance -Target "OU=Windows 10,OU=Computer Accounts,DC=zroots,DC=local"
```

```
Name: windows 10 | ContainerType: OU | Path: ou=windows 10,ou=computer
accounts,dc=zroots,dc=local
GpoInheritanceBlocked: No | GpoLinks: {Windows 10 Security Policy}
InheritedGpoLinks: {Windows 10 Security Policy, Default Domain Policy}
```

## 11. Windows 10 Security Policy GPO Verification (Test 4)

```
Get-GPO -Name "Windows 10 Security Policy" | Select-Object DisplayName, Id, GpoStatus
```

```
DisplayName: Windows 10 Security Policy
Id: 3a184189-759b-420b-883e-9b67a50c78da
GpoStatus: AllSettingsEnabled
```

> **Technical Qualification:** `AllSettingsEnabled` confirms GPO status on the DC, not client receipt. Recommended follow-up: `gpresult /r` or `gpresult /h C:\Temp\Lab03-GPResult.html` on the Windows client.

## 12. Computer Placement & Accidental Deletion Protection (Tests 5–9)

### Test 5: Verify Windows Computer Accounts

```
Get-ADComputer -Filter * | Where-Object { $_.DistinguishedName -like "*OU=Windows 10*" -or
$_.DistinguishedName -like "*OU=Windows 11*" } | Select-Object Name, DistinguishedName
```

```
WIN10-01   CN=WIN10-01,OU=Windows 10,OU=Computer Accounts,DC=zroots,DC=local
WINDOWS11  CN=WINDOWS11,OU=Windows 11,OU=Computer Accounts,DC=zroots,DC=local
```

### Test 6 & 7: Inspect Domain Controllers OU & DC01 Location

```
Get-ADObject -Identity "OU=Domain Controllers,DC=zroots,DC=local" -Properties objectClass,
ProtectedFromAccidentalDeletion | Format-List Name, ObjectClass, DistinguishedName,
ProtectedFromAccidentalDeletion
```

```
Name: Domain Controllers | ObjectClass: organizationalUnit | ProtectedFromAccidentalDeletion:
False
```

### Test 8 & 9: Enable and Verify Accidental-Deletion Protection

```
Set-ADOrganizationalUnit -Identity "OU=Domain Controllers,DC=zroots,DC=local" -
ProtectedFromAccidentalDeletion $true
```

```
Get-ADOrganizationalUnit -Identity "OU=Domain Controllers,DC=zroots,DC=local" -Properties
ProtectedFromAccidentalDeletion | Format-List Name, DistinguishedName,
ProtectedFromAccidentalDeletion
```

```
Name: Domain Controllers | ProtectedFromAccidentalDeletion: True
```

## 13. Troubleshooting & Technical Notes

- **AD PowerShell Modules Missing:** Ensure RSAT (Remote Server Administration Tools) or AD DS role is installed on the management station. Diagnostic: `Get-Module -ListAvailable ActiveDirectory`.
- **Incorrect Distinguished Names:** Typos in `-SearchBase` or `-Identity` cause object-not-found errors. Always verify DN strings via tab-completion or listing.
- **GPO Not Found:** Verify exact GPO display names using `Get-GPO -All`. Names are case-insensitive but must match precisely.
- **Access Denied:** Modifying attributes like deletion protection requires Domain Administrator or Account Operator privileges. Verify current user context with `whoami /priv`.

## 14. Security Lessons & Best Practices

- **Administrative Structure:** Clean OU designs enforce logical separation and least-privilege delegation.
- **Accidental Deletion Safeguards:** Enabling protection prevents catastrophic administrative errors, though it does not replace robust offline backups.
- **Inheritance Visibility:** Understanding direct vs. inherited GPO links prevents policy conflicts and misconfigured baseline deployments.

## 15. Knowledge Check

1. **What is an AD OU?** A container holding users, groups, and computers for delegation and GPO assignment.
2. **What is a DN?** The fully qualified path of an object in Active Directory (e.g., `CN=User,OU=Staff,DC=domain,DC=local`).
3. **What does ProtectedFromAccidentalDeletion=True do?** Prevents users (even administrators) from deleting the container until unchecked.
4. **What does AllSettingsEnabled mean on a GPO?** Indicates the GPO configuration is active, though client-side application depends on scope and gpupdate.

## 16. Lab Conclusion

Lab 03 successfully demonstrated Active Directory OU architecture, PowerShell inspection pipelines, GPO inheritance auditing, and accidental-deletion remediation. All objectives were achieved and verified on `DC01`.

**Evidence Checklist:**
[✓] 12 Verified OUs documented
[✓] Computer placement verified for WIN10-01 & WINDOWS11
[✓] GPO inheritance audited for Linux and Windows 10 OUs
[✓] Domain Controllers OU protection enabled and confirmed (True)